UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	PRE CALCULUS
COURSE CODE	:	MAT133
EXAMINATION	:	NOVEMBER 2026
TIME	:	2 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the MAT133 mock paper (November 2026).
   Answers are unofficial model answers generated from the printed past-paper question; they are not an official UTM marking scheme.

2. Marking points follow the printed part labels and marks. Verify every model answer against the source page.

3. Four questions: 10 + 10 + 10 + 25 marks; total = 55 marks.

4. Questions: 4. Real recycled questions: 4. Placeholders: 0.
   For local study use only; model answers are unofficial.

5. Every question is a real MAT133 past-paper question; its source paper and page are printed with the answer.

FOR MARKERS' USE ONLY

This marking scheme consists of 5 printed pages

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QUESTION 1 (10 marks)

Source: MAT133-2025-JUL, page(s) 2

Unofficial model answer — AI-generated, not an official UTM marking scheme.

a) Solve the inequality $(5-x)^2 \le 9$ and express the solution in real number line.
(5 marks)

Answer: $x \in [2, 8]$; on the real number line, place closed circles at $x = 2$ and $x = 8$ and shade the interval between them.

- $(5-x)^2 \le 9$
- $|5-x| \le 3$
- $-3 \le 5-x \le 3$
- $-8 \le -x \le -2$
- $2 \le x \le 8$

b) Write the equation of circle $x^2+y^2-4x-2y-11=0$ in standard form. Hence, sketch the circle with the correct center and radius.
(5 marks)

Answer: Standard form: $(x-2)^2+(y-1)^2=16$. Center: $(2,1)$; radius: 4. Sketch a circle centered at $(2,1)$ with radius 4, passing through $(6,1)$, $(-2,1)$, $(2,5)$ and $(2,-3)$.

- $x^2+y^2-4x-2y-11=0$
- $(x^2-4x+4)+(y^2-2y+1)=11+4+1$
- $(x-2)^2+(y-1)^2=16$

• Center $(2,1)$, radius $r=\sqrt{16}=4$

Model confidence: high.

QUESTION 2 (10 marks)

Source: MAT133-2022-FEB, page(s) 2

Unofficial model answer — AI-generated, not an official UTM marking scheme.

a) Solve the inequality $\frac{2}{-2x-1} \le 2$. Express the answer in real number line.
(5 marks)

Answer: $(-\infty, -1] \cup \left(-\frac{1}{2}, \infty\right)$

- $\frac{2}{-2x-1} - 2 \le 0$
- $\frac{2-2(-2x-1)}{-2x-1} = \frac{4(x+1)}{-2x-1} \le 0$

• Critical values: $x=-1$ and $x=-\frac{1}{2}$

• Sign table gives negative on $(-\infty, -1)$ and $(-\frac{1}{2}, \infty)$, positive on $(-1, -\frac{1}{2})$

• Include $x=-1$ since numerator is zero; exclude $x=-\frac{1}{2}$ since denominator is zero

• Solution: $(-\infty, -1] \cup \left(-\frac{1}{2}, \infty\right)$

b) If $Z_1 = 2-7i$ and $Z_2 = -2+3i$, state $\frac{\overline{Z}_1 i^{23}}{Z_2}$ in standard form $a+bi$.
(5 marks)

Answer: $-\frac{20}{13} - \frac{17}{13}i$

- $i^{23} = i^{20+3} = i^3 = -i$

- $\overline{Z_1} = 2 + 7i$
- $(2 + 7i)(-i) = 7 - 2i$
- $\frac{7-2i}{-2+3i} \cdot \frac{-2-3i}{-2-3i} = \frac{-20-17i}{13}$
- $= \frac{-20}{13} - \frac{17}{13}i$

Model confidence: high.

QUESTION 3 **(10 marks)**

Source: MAT133-2026-FEB, page(s) 2

Unofficial model answer — AI-generated, not an official UTM marking scheme.

a) Given that $p(x) = \frac{7}{x} + 10$ and $q(x) = \sqrt{x-5}$.

i) State the domain of $p(x)$ and $q(x)$.
(2 marks)

ii) Determine $(q \circ p)(x)$.
(3 marks)

Answer: $\text{Dom}(p) = (-\infty, 0) \cup (0, \infty)$; $\text{Dom}(q) = [5, \infty)$ | $(q \circ p)(x) = \sqrt{\frac{7}{x} + 5} = \sqrt{\frac{5x+7}{x}}$, with domain $(-\infty, -\frac{7}{5}] \cup (0, \infty)$

- $p(x) = \frac{7}{x} + 10$ requires $x \neq 0$
- $q(x) = \sqrt{x-5}$ requires $x - 5 \geq 0 \rightarrow x \geq 5$
- $(q \circ p)(x) = q(p(x)) = \sqrt{\frac{7}{x} + 10 - 5}$
- $= \sqrt{\frac{7}{x} + 5} = \sqrt{\frac{5x+7}{x}}$
- $x \neq 0$ and $\frac{5x+7}{x} \geq 0 \rightarrow x \in (-\infty, -\frac{7}{5}] \cup (0, \infty)$

b) Given that $f(x) = \frac{4-3x}{5x-7}$, find the value of x if $f^{-1}(x) = 2$.
(5 marks)

Answer: $x = -\frac{2}{3}$

- $f^{-1}(x) = \frac{4+7x}{5x+3}$
- $\frac{4+7x}{5x+3} = 2$
- $4 + 7x = 10x + 6$
- $3x = -2$
- $x = -\frac{2}{3}$

Assumptions: The domain of the composite function is stated in addition to the expression, although the subpart only asks to determine $(q \circ p)(x)$.

Model confidence: high.

QUESTION 4 **(25 marks)**

Source: MAT133-2019-JUN, page(s) 3

Unofficial model answer — AI-generated, not an official UTM marking scheme.

a) Given $f(x) = \sqrt{x-9}$, $x \geq h$ and $g(x) = \frac{3-2x}{5+x}$, $x \neq k$. Find

i) the values of h and k
(2 marks)
ii) $(f \circ g)(x)$
(3 marks)
iii) $g^{-1}(2)$
(4 marks)

Answer: i) $h = 9,\ k = -5$; ii) $(f \circ g)(x) = \sqrt{\frac{-11x-42}{5+x}}$; iii) $g^{-1}(2) = -\frac{7}{4}$

- f requires $x - 9 \geq 0 \rightarrow x \geq 9$, so $h = 9$.
- g requires $5 + x \neq 0 \rightarrow x \neq -5$, so $k = -5$.
- $(f \circ g)(x) = f(g(x)) = \sqrt{g(x) - 9}$.
- $g(x) - 9 = \frac{3-2x}{5+x} - 9 = \frac{3-2x-45-9x}{5+x} = \frac{-11x-42}{x+5}$.
- Need $g(x) \geq 9$, giving $-5 < x \leq -\frac{42}{11}$.
- Set $\frac{3-2x}{5+x} = 2$.
- $3 - 2x = 2(5 + x) = 10 + 2x$.
- $-7 = 4x \rightarrow x = -\frac{7}{4}$.

b) Use long division to find the quotient and remainder when $P(x) = 2x^3 - x$ is divided by $(x - 3)$.
(5 marks)

Answer: Quotient $= 2x^2 + 6x + 17$; remainder $= 51$.

- $2x^3 + 0x^2 - x + 0 \div (x - 3)$.
- $2x^2(x - 3) = 2x^3 - 6x^2$, remainder $6x^2 - x$.
- $6x(x - 3) = 6x^2 - 18x$, remainder $17x$.
- $17(x - 3) = 17x - 51$, remainder 51.
- $P(x) = (x - 3)(2x^2 + 6x + 17) + 51$.

c) Solve the equation $5^{x+1} - 5^{2x} = 4$.
(6 marks)

Answer: $x = 0$ or $x = \log_5 4$

- $5^{x+1} - 5^{2x} = 4 \rightarrow 5 \cdot 5^x - (5^x)^2 = 4$.
- Let $y = 5^x > 0$: $5y - y^2 = 4 \rightarrow y^2 - 5y + 4 = 0$.
- $(y - 1)(y - 4) = 0 \rightarrow y = 1$ or $y = 4$.
- $5^x = 1 \rightarrow x = 0$; $5^x = 4 \rightarrow x = \log_5 4$.

d) Sketch the graph of $y = |x|$. Hence, sketch the graph of $y = 5 - |x - 5|$ by using transformation techniques.
(5 marks)

Answer: Start with $y = |x|$ (V-shape, vertex $(0, 0)$). Then $y = |x - 5|$ shifts right 5; $y = -|x - 5|$ reflects in the x-axis; $y = 5 - |x - 5|$ shifts up 5. The final graph is an inverted V with vertex $(5, 5)$, intercepts $(0, 0)$ and $(10, 0)$, and piecewise $y = x$ for $x \leq 5$, $y = 10 - x$ for $x \geq 5$.

- $y = |x|$: vertex $(0, 0)$, slopes ± 1.
- $y = |x - 5|$: shift right 5, vertex $(5, 0)$.
- $y = -|x - 5|$: reflect in the x-axis.

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- $y = 5 - |x - 5|$: shift up 5, vertex $(5, 5)$; intercepts $(0, 0), (10, 0)$.

Assumptions: Take h and k as the lower endpoint and excluded value of the maximal real domains of f and g respectively.
Model confidence: high.

END OF MARKING SCHEME

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